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BCBCA
CCDDE
CDDAE
DEECC
DCBEB
ADCD

Levels of Cleanup	Total Cost of Cleanup ($)	Total Benefit of Cleanup ($)
0	0	0
1	7	45
2	37	80
3	92	105
4	172	125
5	272	140

The table above shows the total cost and the total benefit of cleaning up pollution in a community.

1. Which of the following cleanup levels is socially optimal?

(A) 1

(B) 2 ✓

(C) 3

(D) 4

(E) 5

Answer B

Correct. The total cost is rising with cleaning up pollution, while the total benefit is decreasing. Clean up efforts should be undertaken as long as the marginal benefit of additional cleanup exceeds the marginal cost. The optimal level of cleanup is 2 because the marginal benefit of the second cleanup ($35) exceeds the marginal cost ($30). Beginning with the third level of cleanup, the marginal cost ($55) exceeds the marginal benefit ($25). Thus, the optimal level is 2.

Assume the following utility schedule for Aaliyah's ice cream consumption.

Quantity Consumed	Marginal Utility
1	8
2	6
3	4
4	2
5	0
6	-2

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2. If she consumes 6 units, what is the value of her total utility from consuming ice cream?

(A) 22
(B) 20
(C) 18 ✓
(D) 0
(E) -2

---

3. According to the law of diminishing marginal utility, which of the following is true?

(A) Total satisfaction decreases as more units of a good are consumed.
(B) The additional satisfaction received from consuming extra units of a good decreases as consumption of the good increases. ✓
(C) The additional satisfaction received from consuming extra units of a good decreases as consumption of the good decreases.
(D) The additional satisfaction received from consuming extra units of a good increases as consumption of the good increases.
(E) When marginal utility is decreasing, total utility is decreasing.

4. According to the theory of consumer behavior, which of the following decreases first as additional units of a product are consumed?

(A) Total utility
(B) Average utility
(C) Marginal utility ✓
(D) Marginal physical product
(E) Total physical product

5. The additional satisfaction received from consuming an additional unit of a good is called the

(A) marginal utility ✓
(B) marginal product
(C) average product
(D) average utility
(E) total utility

An art student spends all of her income on pencils and drawing pads. The student currently buys 30 pencils at $1 each and 10 pads at $5 each.

6. If the marginal utility of the 30th pencil is 100 utils and the marginal utility of the 10th pad is 400 utils, to maximize utility the student should change her purchases of pencils and pads in which of the following ways?

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(A)

Pencils	Pads
Not change	Buy more

(B)

Pencils	Pads
Buy more	Buy more

(C) ✓

Pencils	Pads
Buy more	Buy less

(D)

Pencils	Pads
Buy less	Buy more

(E)

Pencils	Pads
Buy less	Buy less

7.

	Apples	Bananas
Marginal utility of last unit	15	12
Price per unit	$5	$3

The table provided shows Ricardo's marginal utility of the last unit consumed of apples and bananas and the price of each good. Ricardo has $125 to spend on apples and bananas. He currently buys 10 apples and 25 bananas. Assuming that he remains within his budget constraint, Ricardo should do which of the following to increase his utility?

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(A) Buy more of both goods
(B) Buy fewer of both goods
(C) Buy more bananas and fewer apples ✓
(D) Buy more apples and fewer bananas
(E) Continue buying the same quantities of both goods

Answer C

Correct. At the initial consumption level, the marginal utility per dollar from consuming apples is $\frac{15}{\$5} = 3$, which is less than $\frac{12}{\$3} = 4$ for bananas. Thus, Ricardo has the incentive to buy more bananas and fewer apples because he gets more marginal utility per dollar for bananas than apples.

8. Assume Pat spends all of her allowance to purchase 4 apples and 4 candy bars. Pat's marginal utility of the fourth apple is 20 utils, and her marginal utility of the fourth candy bar is 40 utils. If an apple costs $1.00 and a candy bar $0.50, to maximize utility Pat should

(A) purchase more apples and more candy bars
(B) maintain the current purchase of 4 apples and 4 candy bars
(C) purchase more apples and fewer candy bars
(D) purchase fewer apples and more candy bars ✓
(E) purchase fewer apples and fewer candy bars

9. Assume that a consumer spends all her income on the purchase of two goods. If the consumer's income doubles and the prices of the two goods also double, the quantity of the two goods purchased will

(A) also double
(B) more than double
(C) increase, but it will be less than double
(D) not change ✓
(E) depend on the slope of the demand curve

10. Assume that Clark spends his entire income on the purchase of two goods, X and Y. If his income and the prices of good X and Y all double, Clark will

(A) double the purchase of goods X and Y
(B) buy more of good X and less of good Y
(C) buy less of good X and more of good Y
(D) buy less of both goods X and Y
(E) buy the same amounts of goods X and Y ✓

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11. At her current level of consumption, a consumer is willing to pay up to $1.50 for a bottle of water and up to $1,500 for a diamond ring because the

(A) total utility of diamond rings is greater than the total utility of water

(B) total utility of water is less than the marginal utility of a diamond ring

(C) marginal utility of a bottle of water is less than the marginal utility of a diamond ring ✓

(D) marginal utility of a bottle of water is greater than the marginal utility of a diamond ring

(E) consumer is irrational and does not understand that water is more important than a diamond

12. Which of the following is true if consuming one unit of a good yields 100 utils and consuming the second unit of the good increases satisfaction by 20 utils?

(A) The marginal utility of the first unit is 20.

(B) The marginal utility of the second unit is 80.

(C) The marginal utility of the second unit is 120.

(D) The total utility of consuming two units is 120. ✓

(E) The total utility of consuming one unit is greater than the total utility of consuming two units.

13. Carlos has a van with 20 seats and charges $10 per person per ride to the airport from downtown. Carlos' cost of the trip is $140 for any number of passengers. On one trip, Carlos has 19 seats filled when a person offers him $5 for the last seat. Should Carlos accept the offer?

(A) No, since the $5 fare offered is below his average cost of $7.

(B) No, since the average variable cost is greater than $5.

(C) No, since it is illegal to charge different prices for the same service.

(D) Yes, since the marginal benefit exceeds the marginal cost. ✓

(E) Yes, since his total revenue exceeds his total cost by $5.

14. If a good is available free of charge, an individual will consume it until

(A) marginal utility is zero ✓

(B) average utility is zero

(C) total utility is zero

(D) marginal utility equals average utility

(E) marginal utility equals total utility

15. Marginal analysis suggests that an individual will consume one additional unit of a good if the

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(A) sunk cost can be recovered
(B) total benefit is less than the total cost
(C) total benefit is greater than the total cost
(D) additional benefit is less than the additional cost
(E) additional benefit is greater than the additional cost ✓

16. Suppose that a consumer purchases two goods X and Y and that the marginal utility of X is MU_x, the total utility of X is TU_x, the marginal utility of Y is MU_y, and the total utility of Y is TU_y. If the prices of X and Y are P_x and P_y, respectively, which of the following expressions defines consumer equilibrium?

(A) $TU_x = TU_y$
(B) $MU_x = MU_y$
(C) $TU_x/P_x = TU_y/P_y$
(D) $MU_x/P_x = MU_y/P_y$ ✓
(E) $(MU_x)(P_x) + (MU_y)(P_y) = 1$

17. To maximize utility, a consumer with a fixed budget will purchase quantites of goods so that the ratios of the marginal utility of each good to its

(A) total utility are the greatest
(B) total utility are the same
(C) price are the greatest
(D) price are equal to one
(E) price are equal ✓

18. Which of the following best describes how a consumer maximizes total utility from the consumption of a bundle of goods and services?

(A) By choosing the quantity of each good such that the quantity demanded of each good is equal to the quantity supplied
(B) By choosing the quantity of each good such that the marginal utility from each good is equal to zero
(C) By choosing the quantity of each good such that the price is equal to the marginal revenue
(D) By choosing the level of output where marginal revenue is equal to marginal cost
(E) By choosing the combination of goods such that the marginal utility per dollar spent on the last unit of each good is equal ✓

Answer E

Correct. Given limited income and the prices of the goods, a consumer maximizes utility by choosing the combination of the two goods such that the marginal utility of the last dollar spent on one good is equal to the marginal utility of the last dollar spent on the other good. That is, the extra utility per dollar spent

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must be the same for the last unit of each good.

19. In a given time period, a person consumes more and more of a good or service and, as a result, enjoys each additional unit less and is willing to pay less for each additional unit. This behavior is consistent with the law of

(A) diminishing returns
(B) diminishing marginal product
(C) diminishing marginal utility ✓
(D) increasing costs
(E) scarce resources

20. For a rational consumer with diminishing marginal utility, the consumption of an additional unit of a good will result in which of the following?

(A) Total utility will increase until it equals marginal utility.
(B) Total utility will increase while marginal utility will be negative.
(C) Total utility will increase while marginal utility will decrease. ✓
(D) Total utility will decrease while marginal utility will be positive.
(E) Total utility will decrease while marginal utility will increase.

Answer C

Correct. Rational consumers with diminishing marginal utility will consume an additional unit of a good as long as the marginal utility of the last unit consumed is positive even if that last unit provides less additional utility than previous units of the good consumed. This happens if the additional utility provided by the consumption of that unit is positive, which will increase total utility.

21. Garcia is currently spending his entire lunch budget on 3 sodas and 4 hot dogs. At his current level of consumption, Garcia's marginal utility for sodas is 5 utils and his marginal utility for hot dogs is 10 utils. In order to maximize his total utility, Garcia should

(A) consume more sodas and fewer hot dogs regardless of the prices
(B) consume more hot dogs and fewer sodas regardless of the prices
(C) maintain his current level of consumption of sodas and hot dogs regardless of the prices
(D) maintain his current level of consumption if the price of a soda is $1 and the price of a hot dog is $2 ✓
(E) maintain his current level of consumption if the price of a soda is $2 and the price of a hot dog is $1

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22. Assume a consumer is spending all her income on two goods: X and Y. At the current consumption combination of the two goods, if the marginal utility per dollar spent on the last unit of good X exceeds that of the marginal utility per dollar spent on the last unit of good Y, what should the consumer do to maximize utility?

(A) Purchase more units of X and more units of Y.

(B) Decrease the price of good X.

(C) Purchase more units of X and less units of Y. ✓

(D) Increase income to buy more of both good Y and good X.

(E) Nothing; utility is maximized at current consumption bundles.

Answer C

Correct. To maximize utility, consumers must allocate their income so that the marginal utility per dollar spent on the last unit of each good is equal. With the marginal utility per dollar of good X greater than that of good Y, the consumer should take a dollar that is being spent on Y and use it to buy a dollar more of X. The consumer should continue the substitution of X for Y until both goods provide the same marginal utility per dollar spent. Doing so will raise total utility.

The table below shows the per-unit prices and marginal utility for the last unit of video games and comic books that Kyle purchased.

	Video Games	Comic Books
Price per unit	$25	$10
Marginal utility	50	50

Kyle spent all of his allocated budget on video games and comic books.

23. To maximize his utility, Kyle should have purchased

(A) more video games and fewer comic books

(B) fewer video games and more comic books ✓

(C) fewer of both goods

(D) equal amounts of both goods

(E) more of both goods

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Answer B

Correct. Given his limited budget, Kyle should consume the combination of video games and comic books such that the marginal utility (MU) of the last dollar spent on video games is equal to the MU of the last dollar spent on comic books. The MU per dollar spent on comic books $(50/\$10)$ is greater than the MU per dollar spent on video games $(50/\$25)$. Therefore, to maximize his utility, Kyle should have bought more comic books and fewer video games.

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24. The information below shows the marginal utilities a student receives from the consumption of pizza and pens. Assume that the marginal utility of each good is independent of the quantity of the other good consumed.

Units of Pizza	Marginal Utility of Pizza	Units of Pens	Marginal Utility of Pens
1	20	1	8
2	18	2	7
3	16	3	6
4	14	4	5
5	12	5	4
6	10	6	3
7	9	7	2
8	8	8	1

If the student purchases 2 units of pizza and 2 units of pens, the student's total utility will be

(A) 4 utils
(B) 25 utils
(C) 28 utils
(D) 50 utils
(E) 53 utils ✓

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25.

Goods	Price	Quantity Purchased (units)	Marginal Utility (utils)
Soda	$1	10	10
Apples	$2	5	25

Jane spends all her weekly allowance to buy only two goods: soda and apples. According to the table above, if her preferences are characterized by the law of diminishing marginal utility, then which of the following statements is correct?

(A) Jane is maximizing her utility.

(B) Jane can buy more apples and less soda to maximize her utility. ✓

(C) Jane can buy more soda and fewer apples to maximize her utility.

(D) Jane can buy more apples and the same amount of soda to maximize her utility.

(E) Jane can buy more soda and the same amount of apples to maximize her utility.

26. The following chart shows the total utility that Juan receives from consuming various amounts of chocolate candy bars each day.

Number of Candy Bars Consumed per Day	Total Utility
0	0
1	40
2	70
3	90
4	100
5	80

Which of the following statements about Juan's marginal utility is correct?

(A) His marginal utility from the first candy bar is greater than his marginal utility from the second candy bar. ✓

(B) His marginal utility from the fourth candy bar is greater than his marginal utility from the third candy bar.

(C) His marginal utility increases at a constant rate.

(D) He first experiences diminishing marginal utility with the consumption of the fifth candy bar.

(E) His greatest marginal utility comes from his consumption of the fourth candy bar.

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27. Let MU_s be the marginal utility of a sandwich, MU_h be the marginal utility of a hot dog, P_s be the price of a sandwich, and P_h be the price of a hot dog. When the price of the goods is zero, Pat eats a sandwich. When Pat has to pay, she eats a hot dog. When Pat has to pay, which of the following is necessarily true?

(A) $MU_s = MU_h = P_s = P_h$

(B) $(MU_s / P_s) > (MU_h / P_h)$

(C) $P_s < P_h$

(D) $P_s > P_h$ ✓

(E) $MU_s = 0$

Number of Apples	Total Utility
2	130
3	180
4	220
5	250
6	270
7	280

28. The marginal utility per dollar spent on the last orange consumed is 75. If the price of an apple is \$0.50, how many apples would Johnny have to consume before he considers purchasing another orange?

(A) 2

(B) 3

(C) 4 ✓

(D) 5

(E) 6

Answer C

Correct. The marginal utility per dollar spent on the fourth apple is $\frac{(220-180)}{\$0.50} = 80$. This is the last apple that has a marginal utility per dollar spent that is greater than 75.

29. If the price of an apple is \$0.50, the marginal utility per dollar spent on the fifth apple is:

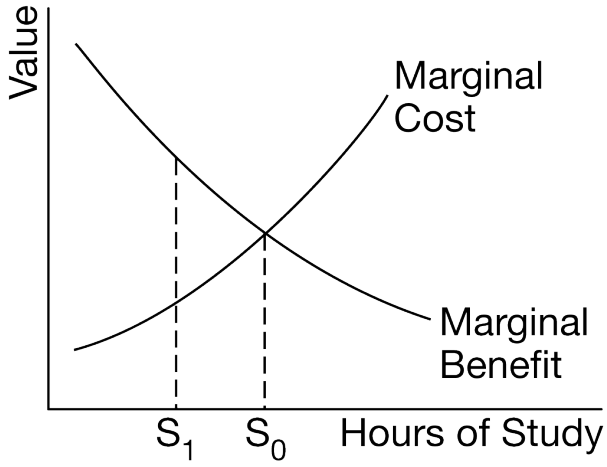
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(A) 20
(B) 30
(C) 40
(D) 60 ✓
(E) 100

Answer D

Correct. The marginal utility per dollar spent on the fifth apple is $\frac{250-220}{0.50} = 60$.

30.



The graph above shows the marginal benefit that a student receives from additional hours of study, and the marginal cost the student will incur from additional hours of study. Which of the following is true?

(A) Beyond S_0 hours of study, there is no further benefit from devoting additional hours to study.
(B) At S_1 hours of study, the extra cost of study time exceeds the extra benefit of study time.
(C) The optimal number of hours that the student should study is S_0 hours. ✓
(D) The rational individual should study more than S_0 hours.
(E) The marginal cost of studying decreases as more time is spent studying.

31. The marginal benefit of consuming a good is

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(A) the change in average utility that results from consuming one more unit of the good
(B) the same as the total benefit
(C) equal to the marginal cost of the good
(D) the change in total expenditures as a result of buying one more unit of the good
(E) the maximum amount a consumer is willing to pay for one more unit of the good ✓

Answer E

Correct. A consumer's willingness to pay reflects how much the consumer values the good or service. Specifically, the demand curve reflects the marginal value of each unit of the good, which is equal to the maximum amount the consumer is willing to pay for that unit. The marginal value decreases as quantity increases.

32. When marginal utility from consuming a good is zero, the total utility from consuming the good is

(A) zero
(B) negative
(C) maximized ✓
(D) rising at a constant rate
(E) equal to marginal utility per dollar spent on the good

Answer C

Correct. Marginal utility represents the change in total utility as a result of consuming one more unit of a good. Once the consumption of a good has reached the point where the marginal utility is zero, additional consumption of the good will result in negative marginal utility and thus a reduction in total utility. Therefore, when marginal utility from consuming a good is zero, total utility has been maximized.

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33.

Milkshake Cups	Total Utility (utils)
1	50
2	100
3	145
4	185
5	220

Which of the following is true based on Rosco's utility data given in the table provided?

(A) Rosco's marginal utility from the fourth cup is 185 utils.

(B) Rosco's marginal utility from the fifth cup is 35 utils. ✓

(C) Rosco's marginal utility from the third cup is greater than the marginal utility from the second cup.

(D) Rosco's marginal utility from the fourth cup is greater than the marginal utility from the third cup.

(E) Rosco's total utility and marginal utility both increase as he consumes more cups of milkshake.

Answer B

Correct. Marginal utility is calculated as the change in total utility/change in units consumed, which is $\frac{220-185}{5-4} = 35$ utils in this case.

34. Assume that Larry spends his entire allowance on the purchase of pretzels and ice-cream cones. The price of a bag of pretzels is $\$1$, and the price of an ice-cream cone is $\$2$. At his current consumption combination, the marginal utility of pretzels is 20 and of ice-cream cones is 60. To maximize his utility, Larry should

(A) buy more of both goods

(B) buy less of both goods

(C) buy more pretzels and fewer ice-cream cones

(D) buy fewer pretzels and more ice-cream cones ✓

(E) maintain his current consumption level

Answer D

Correct. Utility maximization occurs when the marginal utility per dollar of each good is equal. Since the marginal utility per dollar of ice-cream cones $\left(\frac{60}{2} = 30\right)$ exceeds the marginal utility per dollar of pretzels $\left(\frac{20}{1} = 20\right)$, Larry should buy more ice-cream cones and fewer pretzels to maximize his utility.

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Mickey has a $20 gift card that she is planning to use to download 10 songs and 5 games. The download price for each song is $1 and the download price for each game is $2. Her marginal utility from the tenth song is 30, and her marginal utility from the fifth game is 70.

35. Based on this information, which of the following statements is true?

(A) Mickey's plan will maximize her utility.

(B) Mickey can increase utility by buying more songs and fewer games.

(C) Mickey can increase utility by buying more games and fewer songs. ✓

(D) Since games are twice as expensive as songs, Mickey should buy twice as many songs.

(E) Since games are twice as expensive as songs, Mickey should buy twice as many games.

36. When marginal utility is falling but positive, total utility will

(A) decrease at an increasing rate

(B) decrease at a decreasing rate

(C) remain constant

(D) increase at an increasing rate

(E) increase at a decreasing rate ✓

37. The city council divides a community's residents into three groups: individual young adults, families with children, and older adults. The following table summarizes how much each group is willing to pay for each playground.

	Amount Groups Are Willing to Pay for Each Playground		
Number of Playgrounds	Individual Young Adults	Families with Children	Older Adults
1	$400	$2,000	$1,000
2	$300	$1,800	$900
3	$200	$1,600	$800
4	$100	$1,400	$700
5	$0	$1,200	$600
6	$0	$1,000	$500
7	$0	$900	$400
8	$0	$800	$300

The city council must pay $2,250 to build each playground. Which of the following is a characteristic of playgrounds and what is the optimal number of playgrounds for the township to build?

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(A) Playgrounds are nonrival in consumption, and the optimal number of playgrounds is zero.
(B) Playgrounds are nonrival in consumption, and the optimal number of playgrounds is two.
(C) Playgrounds are rival in consumption, and the optimal number of playgrounds is three. ✓
(D) Playgrounds are nonexcludable, and the optimal number of playgrounds is zero.
(E) Playgrounds are excludable in consumption, and the optimal number of playgrounds is two.

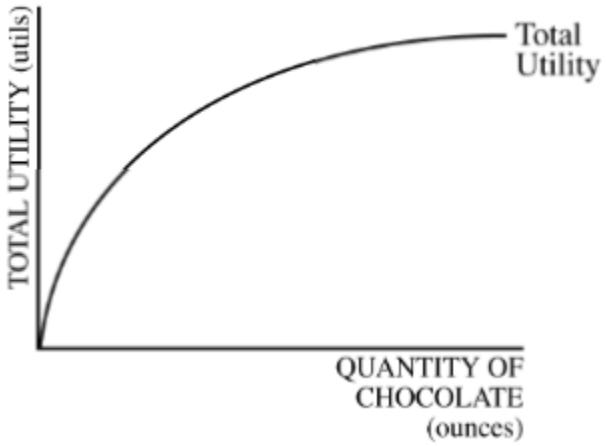
Answer C

Correct. Playgrounds are rival in consumption. As more children use a playground, the enjoyment each child receives is likely to decline due to overcrowding. The optimal number of playgrounds for the community to build is three playgrounds. The community is willing to pay $3,400 for the first playground, $3,000 for the second playground, and $2,600 for the third playground. Each playground will cost $2,250 to build. The first three playgrounds should be built, since the community is willing to pay more for those playgrounds than it will cost to build them. The community is willing to pay only $2,200 for the fourth playground, which is less than the $2,250 cost to build the fourth playground. Therefore, the fourth, and similarly fifth through eighth playgrounds, should not be built.

38. Oscar spent his entire income on only two goods: good X and good Y. At his current consumption of the two goods, the marginal utility of X is 8 and the marginal utility of Y is 2. If the price of X is $4.00 and the price of Y is $0.50, then to maximize his total utility, Oscar should have

(A) bought more X and more Y
(B) bought more X and less Y
(C) bought less X and more Y ✓
(D) bought less X and less Y
(E) maintained his current consumption

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The diagram above shows Sally's utility function for chocolate.

39. Her utility function illustrates the principle of

(A) trade-offs when making choices

(B) diminishing marginal utility ✓

(C) income and substitution effects

(D) increasing opportunity cost

(E) constrained utility maximization

Quantity of Apples	Total Benefit ($)
1	10
2	18
3	24
4	28
5	30
6	31

The table above shows the total benefit Tony gets from eating different quantities of apples.

40. If apples cost $3 each, how many should he buy to maximize his consumer surplus?

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(A) 1
(B) 2
(C) 3
(D) 4 ✓
(E) 5

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41. When total utility is at its maximum, marginal utility is

(A) increasing
(B) negative
(C) equal to zero ✓
(D) at a maximum
(E) at minimum

42. Suppose that Habib has a weekly fixed budget and spends it all on music downloads and snacks. At his current combination of consumption, the marginal utility of the last dollar spent on music downloads is greater than the marginal utility of the last dollar spent on snacks. Has Habib maximized his utility?

(A) Yes, because he has purchased the maximum possible with his limited budget.
(B) Yes, because he has purchased the two goods in proportion so that he can get the maximum utility from each.
(C) Yes, because changing his current consumption combination will reduce his total utility.
(D) No, because he can increase his total utility by purchasing more music downloads and fewer snacks. ✓
(E) No, because he can increase his total utility by purchasing fewer music downloads and more snacks.

Answer D

Correct. The utility-maximizing condition is not fulfilled. Utility maximization requires that the marginal utility of the last dollar spent on music downloads must equal the marginal utility of the last dollar spent on snacks. He can maximize his utility by purchasing more music downloads and fewer snacks. Purchasing more music downloads will reduce the marginal utility of music downloads, decreasing the marginal utility per dollar, and reducing the number of snacks increases the marginal utility of snacks, increasing the marginal utility per dollar; the net impact will be to increase total utility.

43. Josephine is buying candy and gum. If she is maximizing her utility for the combination of candy and gum she buys, which of the following must hold?

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(A) Marginal utility of candy ÷ price of candy = marginal utility of gum ÷ price of gum ✓

(B) Total utility of candy ÷ price of candy = total utility of gum ÷ price of gum

(C) Marginal utility of candy ÷ total utility of candy = marginal utility of gum ÷ total utility of gum

(D) Marginal utility of candy × price of candy = marginal utility of gum × price of gum

(E) Total utility of candy × price of candy = total utility of gum × price of gum

Answer A

Correct. To maximize utility, Josephine must allocate her income so that the marginal utility per dollar spent on the last unit of each good is equal. The formula for maximizing utility is $\frac{\text{MU of candy}}{\text{Price of candy}} = \frac{\text{MU of gum}}{\text{Price of gum}}$.

44. Which of the following is a key assumption of consumer choice theory?

(A) Consumers buy goods as long as implicit costs are greater than explicit costs.

(B) Consumers maximize their utility. ✓

(C) As income rises, consumers buy more of all goods.

(D) Marginal utility is always increasing.

(E) As more inputs are used, consumers get less and less additional output.

45. Assume the price of a candy bar is $2 and the price of a bag of chips is $3. Assume Lilly's marginal utility from consuming an additional candy bar is 10 utils and her marginal utility from consuming an additional bag of chips is 12 utils. If Lilly is spending her fixed weekly allowance on candy bars and bags of chips, which of the following actions will maximize Lilly's total utility?

(A) Buying more candy bars and bags of chips

(B) Buying more candy bars and fewer bags of chips ✓

(C) Buying fewer candy bars and more bags of chips

(D) Buying fewer candy bars and bags of chips

(E) Buying the same number of candy bars and bags of chips

Answer B

Correct. To maximize utility, Lilly must allocate her income so that the marginal utility per dollar spent on the last unit of each good is equal. The marginal utility per dollar spent on the last candy bar is 5 utils per dollar (10 utils divided by $2), and the marginal utility per dollar spent on the last bag of chips is 4 utils per dollar (12 utils divided by $3). Given Lilly's fixed weekly allowance, Lilly should buy more

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candy bars and fewer bags of chips.